

Fine-grained sentiment analysis of catering reviews based on ELECTRA-DPCNN model

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ABSTRACT

Aiming at the core problems of noise interference and difficulty in long text modeling in fine-grained sentiment analysis of catering e-commerce reviews, a hybrid neural network of ELECTRA pre-trained model and deep pyramid convolution, ELECTRA-DPCNN, is proposed. The model innovatively combines ELECTRA's Replaced Token Detection (RTD) task and DPCNN's multi-scale feature abstraction ability. Firstly, ELECTRA is used to replace the traditional BERT for efficient semantic coding, and the domain generalization is strengthened by RTD; Then a hierarchical convolution architecture is designed, and the self-attention mechanism is embedded in DPCNN. The processing flow of "region embedding, narrow convolution, residual convolution block, downsampling, self-attention" is constructed to realize the gradual feature pyramid abstraction. The model significantly optimizes the ability to capture long-distance emotional cues, and self-attention mechanism effectively filters domain independent noise. In the test of Baidu Qianyan ASAP-ASPECT dataset (a large Chinese fine-grained dataset, including 213371 catering reviews), the accuracy of the model is 86.64%, which is significantly better than other models, and provides a new paradigm of high-precision and interpretable emotion analysis for catering e-commerce decision-making.

Keywords: ELECTRA; DPCNN; Self-attention; Catering e-commerce review; Aspect-Based Sentiment Analysis

1. INTRODUCTION

With the expansion of e-commerce platforms to catering and other service industries, users are increasingly inclined to share store reviews on social platforms. These reviews not only enhance consumers' sense of participation, assist consumer decision-making, but also help enterprises optimize service quality. In this context, machine learning driven fine-grained sentiment analysis technology shows key value, covering different dimensions such as word level, sentence level and aspect level analysis. As one of the core technologies in the field of natural language processing, aspect-based sentimental analysis (ABSA) aims to accurately identify specific aspects (such as product features, service objects or events) and their emotional polarity (positive, negative or neutral) from text [1]. In the face of complex expressions such as "clear mobile screen but fast battery power consumption", the traditional coarse-grained analysis has been unable to meet the demand, and the algorithm that can analyze specific aspects of emotion is urgently needed. This technology has important application prospects in the fields of user portrait construction, public opinion monitoring and enterprise decision support.

Under the background of explosive growth of user comments on e-commerce platforms, fine-grained sentiment analysis faces core challenges such as insufficient domain migration ability and noise interference. The existing methods still have some limitations: first, although the pre-trained model enhances semantic understanding, it has insufficient ability to filter domain independent noise (such as non-standard abbreviations or spelling errors in user comments); Second, the traditional neural network is relatively difficult to balance the long-distance dependence modeling and computational efficiency, resulting in the loss of key emotional clues during migration. Therefore, the ELECTRA-DPCNN model is proposed. By using ELECTRA for text vector mapping and feature selection, the trained efficiency is greatly improved while preserving the accuracy. Then it is input into DPCNN combined with self-attention mechanism for progressive feature abstraction, which significantly improves the accuracy of capturing the implicit emotional patterns in long food and beverage reviews.

2. LITERATURE REVIEW

2.1 Research based on emotional dictionary and syntactic rule

Early fine-grained sentiment analysis methods were mainly based on rule driven language processing and shallow machine learning model. By constructing an emotional dictionary and grammatical rules [2], the aspect words and emotional polarity are extracted from the text. However, this approach has significant limitations, such as difficulty in dealing with polysemy and emotional expression in complex context. Support vector machine (SVM) and other shallow machine learning models have alleviated these problems to a certain extent [3]. The text is transformed into structured data for classification through feature engineering, but they still rely heavily on manually designed features and cannot effectively capture the context or diversified expression. Therefore, although the early methods laid a theoretical foundation in the field of emotion analysis, their limitations prompted researchers to turn to deep learning to better cope with the challenges of complex emotion understanding.

2.2 Research based on convolutional neural network

And then models based on convolutional neural network (CNN) show a strong ability of text feature extraction. Text-CNN captures local emotion patterns through convolution kernel of fixed window, providing an efficient solution for short text emotion classification [4]. RCNN combines convolution and loop structure to enhance the modeling ability of context dependence, which is more suitable for long text sentiment analysis [5]. In addition, Gated-CNN introduces a gating mechanism to dynamically adjust the feature extraction process, which can more accurately identify the key areas of emotional information [6]. These models lay an important foundation for capturing diverse semantic features in emotion analysis tasks. In terms of model architecture innovation, the deep pyramid convolution network (DPCNN) gradually expands the receptive field to capture long-distance dependence through hierarchical convolution and pooling operation, which solves the problem of traditional CNN being overly sensitive to local features [7]. For example, the AT-DPCNN model combines the attention mechanism to dynamically weight the emotional key areas in the text. In the classification task of transitional sentences (such as "excellent screen but continuous gap"), F1 value is 4.2% higher than that of traditional CNN [8]. The advantage of DPCNN lies in the balance between its computational efficiency and feature abstraction ability: by reducing the loss of information in the pool layer, its hierarchical structure can gradually refine the semantic pyramid of the text, especially when dealing with multi-objective long text common in e-commerce reviews, it shows stronger robustness.

2.3 Research based on pre-trained language models

With the emergence of attention mechanism and Transformer, the pre-trained language model based on this has changed the research paradigm. Represented by BERT (Bidirectional Encoder Representations from Transformers), it learns context semantic representation from large-scale corpus through masked language modeling (MLM) and next sentence prediction (NSP) tasks, which significantly improves the domain generalization ability of the model [9]. Subsequent research focused on the optimization of BERT: RoBERTa enhanced semantic modeling through dynamic mask strategy and larger scale trained data [10]; ALBERT reduces the number of model parameters to 1/9 of BERT through embedded layer parameter decomposition and cross layer parameter sharing, while maintaining performance [11]; ERNIE introduces knowledge mapping to enhance entity semantic understanding [12]; ELECTRA innovatively adopts the generator-discriminator architecture to replace MLM through replaced token detection (RTD), which improves the pre-trained efficiency by more than three times [13]. In 2024, Microsoft Research Asia further proposed ELECTRA+HP-Dist model to optimize the sampling efficiency of the generator through difficulty prediction and sampling smoothing strategy, and the F1 value in the squad2.0 question and answer task increased by 1.8% [14]. These variant models provide a more efficient semantic coding basis for sentiment analysis.

3. CONSTRUCTION OF SENTIMENT ANALYSIS MODEL BASED ON ELECTRA-DPCNN

3.1 ELECTRA model

ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) is a pre-trained language model proposed in 2020. It is significantly improved on the basis of BERT, which proposes the pre-trained task of "replaced token detection" (RTD) to replace MLM, similar to GAN, and greatly improves the sample efficiency of pre-trained and the performance of the model on downstream tasks.

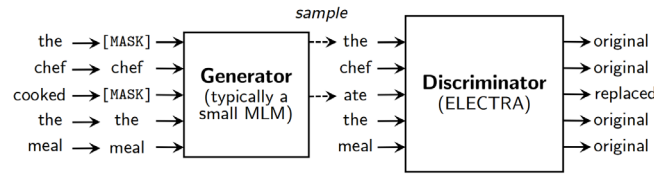


Figure 1. the Structure of ELECTRA.

As shown in the Fig. 1, ELECTRA includes generator and discriminator, both of which adopt the encoder structure of Transformer [15], as shown in the figure below.

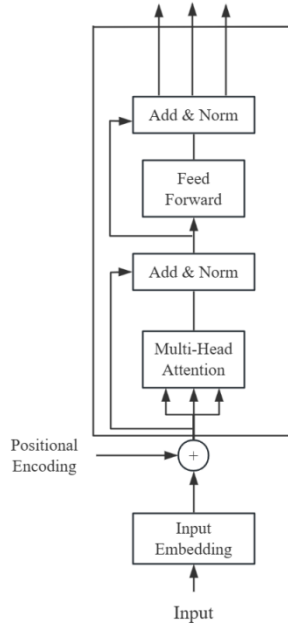


Figure 2. the Structure of transformer encoder.

The generator is a small masked language model. First, 15% of the tokens are randomly selected and replaced by masked tokens. Because ELECTRA uses its discriminator in fine-tuning, it does not need to consider mitigating the mismatch between pre-trained and fine-tuning, and it cancels the operation of 80% masked, 10% unchanged and 10% random replaced of BERT, which greatly improves the trained efficiency. Then, the generator is used to train the model to predict the masked token and get the corrupted token;

As shown in Fig. 1, the generator randomly masks two tokens, namely "the" and "cooked". After the generator's prediction, the corresponding corrupted tokens are obtained. Among them, "the" is successfully predicted, while "cooked" is changed to "ate".

The function of the discriminator is to distinguish whether each input token is original or replaced. If the token made by the generator is consistent with the original token, then this token is still original, such as "the" in the above example. So for each token, the discriminator will perform a binary classification and finally obtain the loss on all tokens.

The trained objective of the generator remains MLM, while the objective of the discriminator is sequence labeling. Both are trained simultaneously, but the gradient of the discriminator is not passed to the generator. Given the input $\mathbf{x} = [x_1, x_2, \dots, x_n]$, MLM first select a random set of positions (integers between 1 and n) to mask out $\mathbf{m} = [m_1, \dots, m_k]$. The objective function is as follows:

$$\min_{\theta_G, \theta_D} \sum_{\mathbf{x} \in \square} \mathcal{L}_{\text{MLM}}(\mathbf{x}, \theta_G) + \lambda \mathcal{L}_{\text{Disc}}(\mathbf{x}, \theta_D) \quad (1)$$

Among them:

$$\mathcal{L}_{\text{MLM}}(\mathbf{x}, \theta_G) = \mathbb{E} \left(\sum_{i \in \mathbf{m}} -\log p_G(x_i | \mathbf{x}^{\text{masked}}) \right) \quad (2)$$

$$\mathcal{L}_{\text{Disc}}(\mathbf{x}, \theta_D) = \mathbb{E} \left(\sum_{t=1}^n -\mathbb{1}(x_t^{\text{Corrupt}} = x_t) \log D(\mathbf{x}^{\text{Corrupt}}, t) - \mathbb{1}(x_t^{\text{Corrupt}} \neq x_t) \log(1 - D(\mathbf{x}^{\text{Corrupt}}, t)) \right). \quad (3)$$

The generator and discriminator improve the model through weight sharing, but due to size reasons, only the embedding weight of the generator is shared, while the generator is trained by MLM, which predicts the token according to the context around the token, and has better learning ability for embedding.

3.2 DPCNN model combined with attention mechanism

DPCNN (deep pyramid convolutional neural networks) is a deep convolutional neural network for text classification task proposed by Rie Johnson et al. in 2017 [16]. By introducing the pyramid downsampling structure and residual connection, the network is deepened while maintaining the computational efficiency, and the feature extraction ability of long text is significantly improved.

The DPCNN, enhanced with an added attention mechanism, is depicted in the figure below and comprises three main components. Firstly, there is the region embedding: it takes a word embedding sequence as input. The obtained output passes through the convolution block, complemented by zero padding at both ends to ensure that the input and output sequences maintain the same length, thereby guaranteeing that the output for each word position encapsulates its surrounding context information. The pre-activation approach is employed, where ReLU activation precedes convolution, facilitating accelerated trained convergence. Subsequently, pyramid downsampling and residual connections are utilized. A maximum pooling layer is inserted after convolutional block, halving the sequence length while preserving the number of feature maps. This process is repeated to form a pyramid structure. Drawing inspiration from ResNet, residual connections are adopted, connecting the input of each convolutional block to its output via shortcuts, thereby addressing the issue of gradient vanishing in deep networks.

In order to improve the local feature extraction ability of the model, a self-attention mechanism is added after the output of the convolution layer, namely:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

Among them:

$$Q = XW^Q \quad (5)$$

$$K = XW^K \quad (6)$$

$$V = XW^V \quad (7)$$

X is the input word vector matrix, W^Q , W^K and W^V are three trainable parameter matrices, d_k is the dimension size of K .

In addition, a narrow convolution layer is respectively added after region embedding and pooling to further enhance fine-grained sentiment analysis.

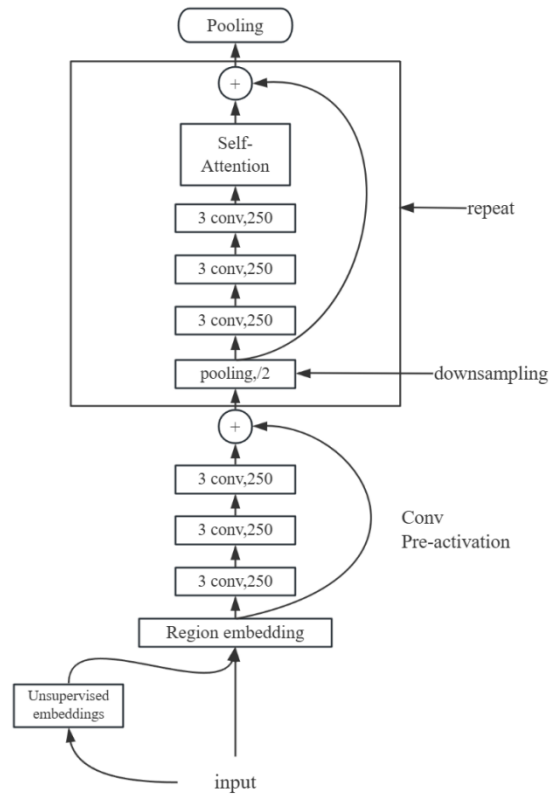


Figure 3. the Structure of DPCNN.

3.3 Construction of ELECTRA-DPCNN model

The model design is shown in the figure below. In general, the new model ELECTRA-DPCNN is obtained by combining ELECTRA model and DPCNN model combined with attention mechanism. ELECTRA-DPCNN consists of three parts: input layer (ELECTRA), DPCNN layer and output layer. That is, the input text is pre-trained with the ELECTRA model, and the last hidden layer is used as the feature extraction vector of the text. These vectors are transposed and input into the DPCNN layer. After multiple convolution, downsampling and self-attention mechanism, the emotional features of the final text are output, and the global semantic information of the text is further strengthened. Finally, the emotional classification is obtained in the output layer through the softmax classifier.

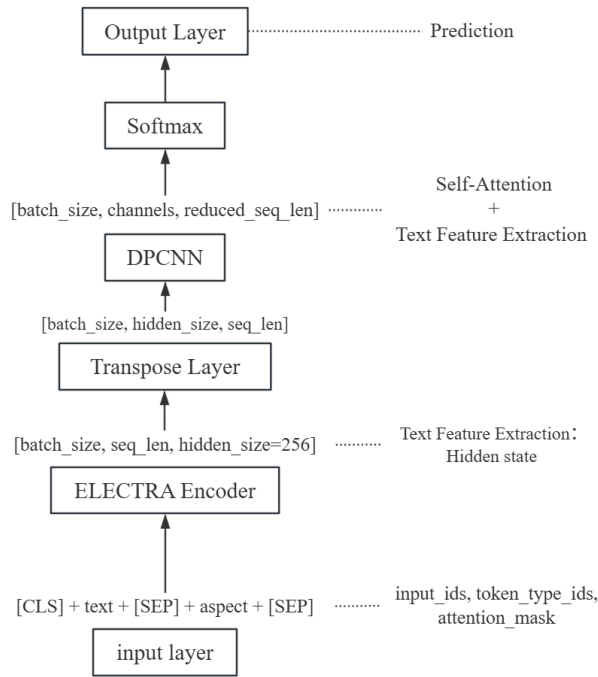


Figure 4. the Structure of ELECTRA-DPCNN.

3.3.1 Input layer (ELECTRA)

The main purpose of this layer is to vectorize the original data, and at get the shallow semantic features of the text meanwhile. The preprocessed text (punctuation cleaned out) is represented as context and aspect words mixed word embedding matrix $([CLS], S_1, S_2, \dots, S_n, [SEP], A_1, A_2, \dots, A_m, [SEP])$, which are added with `input_ids`, `token_type_ids`, `attention_mask`. Among them, `[CLS]` for classification task, $S_1, S_2, \dots, S_n$ for context information, $A_1, A_2, \dots, A_m$ for aspect word information, `[SEP]` for end of sentence.

This matrix is used as the input through the ELECTRA encoder, and the output hidden state $[batch_size, seq_len, hidden_size]$ is used as the shallow text feature dimension to obtain the processed mixed word embedding matrix H in the same form as before.

3.3.2 DPCNN layer combined with self-attention mechanism

This layer is built based on the improved DPCNN network and self-attention mechanism, mainly including region embedding, narrow convolution, convolution blocks, residual connection, downsampling, as well as self-attention. After inputting the dimension transpose, take the mixed matrix H as the input of this layer and perform the following operation:

$$C = f(W \cdot H + b) \quad (8)$$

Where W is the weight matrix, b is the bias, and f is the convolution function. After downsampling, the features in C are further extracted to obtain the feature vector $\hat{C}$. In order to further obtain the semantic information and relevance of the sentence context, the eigenvector is passed through the self-attention mechanism, and finally the optimized eigenmatrix W^E is obtained as the output of DPCNN layer. The output dimension is $[batch_size, channels, reduced_seq_len]$ and enters the next layer.

Through the DPCNN layer, the extracted features can contain the global semantic information of the text, deal with the long-distance dependence and key emotional expression positioning problems in emotional analysis, and finally output the highly abstract sentiment feature representation. In the model methodology section, the integration of predictive modeling with reinforcement learning strategies, as outlined by Yu [18], could enhance the dynamic feature abstraction in ELECTRA-DPCNN by enabling adaptive optimization based on sequential sentiment feedback.

3.3.3 Output layer

Input the obtained feature vector W^E into softmax classifier for emotion classification. The classification method is as follows:

$$y = \text{softmax}(WW^E + b) \quad (9)$$

The loss function is a cross entropy function to optimize the model, as shown in (10):

$$\text{loss} = - \sum_{i \in D} \sum_{j \in C} \hat{y}_i \log(y_i^j) + \lambda \|\theta\|^2 \quad (10)$$

Where D is the trained set, C is the number of categories, $\hat{y}$ is the actual category, y is the predicted category of sentences, and $\lambda \|\theta\|^2$ is the regular term.

4. EXPERIMENTAL DESIGN AND RESULT ANALYSIS

4.1 Experimental data and pretreatment

For this task, the dataset of Baidu thousand words competition based on the Chinese dataset of meituan ASAP-ASPECT was selected for the experiment. In this dataset, there were 213371 development sets, 29101 trained sets and 28363 trained sets. Emotions were divided into three categories: positive (1), neutral (0) and negative (-1). The fine-grained categories covered are shown in the following table.

Table 1. Aspect category.

<i>Aspect category</i>	<i>Aspect category</i>
Food#Taste	Price#Cost_effective
Food#Portion	Price #Discount
Food# Recommend	Ambience#Sanitary
Food#Appearance	Ambience#Space
Service#Queue	Ambience#Decoration
Service#Parking	Ambience#Noise
Service #Timely	Location#Transportation
Service#Hospitality	Location#Easy_to find
Price#Level	Location#Downtown

The emotional distribution of each category is shown in the following figure:

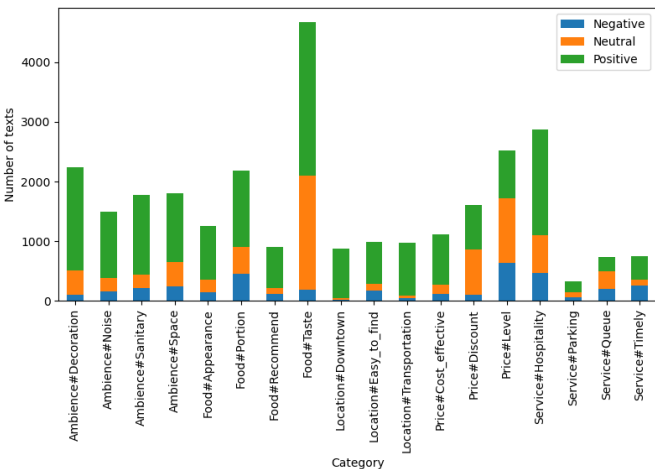


Figure 5. the Distribution of dataset.

And the spaces, punctuation marks, expressions and tabs carried by the data were cleaned.

4.2 Experimental environment and configuration

This experiment was carried out on the baidu AI studio platform. The framework is PaddlePaddle 2.4.0, the python version is 3.7, the GPU is Tesla V100, the Video Mem is 16GB, the RAM is 16GB, and the Disk is 100GB.

In the trained process, AdamW is used as the optimizer combined with weight decay, the learning rate is 2e-5, the L2 regularization parameter is 1e-4, the batch size is 32, the maximum length of the sentence is 128, the epoch is set to 50, and the patient parameter of the early stop mechanism is set to 15. The trained process is shown in the following figure:

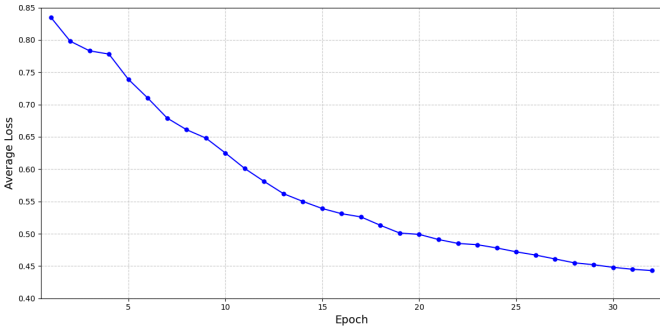


Figure 6. Average loss over epochs.

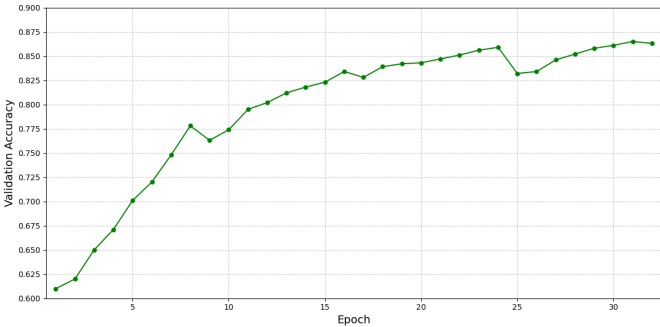


Figure 7. Validation accuracy over epochs.

4.3 Experimental results and comparison

4.3.1 Trained results of different pre-trained models:

To verify the advantages of ELECTRA in this task, this paper selected the baseline Chinese models of different pre-trained models for training, and compared the trained results on the same dataset. The results are shown in the table below:

Table 2. Trained results of different pre-trained models.

<i>Pre-trained model</i>	<i>Precision(%)</i>
BERT	81.13
RoBERTa	82.12
ALBERT	81.53
ERNIE	81.32
ELECTRA	82.30

It can be seen that ELECTRA performs best among different models with an accuracy of 82.30%, which shows advantages of its RTD pre-trained task. Compared with BERT's mask language modeling and RoBERTa's dynamic mask strategy, ELECTRA's RTD realizes more efficient semantic boundary learning by detecting the corrupted tokens replaced by the generator through the discriminator, and obtains richer context representation ability under the same training resources. This architecture makes the model more robust to non-standard expressions in user comments, and significantly improves the sensitivity to emotional keywords, laying a high-quality semantic foundation for further feature extraction. Therefore, ELECTRA model is selected for improvement.

4.3.2 Trained results of different neural networks

To find deep neural networks with better feature extraction ability, this paper trained different networks on this dataset as shown below:

Table 3. Trained results of different Deep neural networks.

<i>Deep neural network</i>	<i>Precision(%)</i>
BiLSTM	71.13
Text-CNN	71.42
DPCNN	73.56
Self-Attention_DPCNN	74.10

It can be found that the accuracy of the three deep networks (BiLSTM,Text-CNN,DPCNN) has subtle difference, but the combination of self-attention mechanism can significantly enhance the text feature extraction ability of the network, and the accuracy on DPCNN network has reached 74.10%.

4.3.3 ELECTRA-DPCNN model trained results

After the ELECTRA-DPCNN model training, the effect is verified and compared with other similar models on the test set. The results are shown below. ELECTRA-DPCNN demonstrates notable advantages over other models, achieving 86.64% accuracy by synergizing ELECTRA's efficient RTD task with DPCNN's multi-scale feature pyramid. This hybrid architecture uniquely enhances noise robustness while capturing nuanced sentiment expressions in long-tail reviews, gaining 82.07% in F1-score.

Table 4. Trained results and comparasions of ELECTRA-DPCNN AND OTHER MODELS.

<i>Model</i>	<i>Precision(%)</i>	<i>F1-score(%)</i>
ERNIE_ BiLSTM	83.82	79.05
ERNIE_DPCNN	84.75	82.83
ALBERT_DPCNN	79.43	79.01
ELECTRA_ Text-CNN	85.21	80.35
ELECTRA_DPCNN	86.64	82.07

5. CONCLUSION

Aiming at the challenge of fine-grained sentiment analysis in the field of catering e-commerce, this paper innovatively integrates ELECTRA pre-trained model and deep pyramid convolutional network (DPCNN) to construct an efficient and accurate analysis framework. Traditional methods relying on models such as BERT cannot fully capture the semantic boundaries and emotional patterns in user comments, while ELECTRA's replaced token detection mechanism significantly improves the efficiency of semantic modeling. Combined with DPCNN's multi-scale feature pyramid structure, it effectively solves the problem of emotional expression analysis in long context. Through verification on Baidu Qianyan ASAP-ASPECT dataset, the model achieved 86.64% accuracy in catering scene, and its self-attention mechanism successfully filtered noise in most fields, providing an interpretable emotion analysis tool for e-commerce catering platform.

The actual deployment shows that the ELECTRA-DPCNN model has obvious advantages over other baseline models and composite models. The current research still has room for improvement in dynamic emotion modeling and cross scene adaptability. In the future, it will strengthen the analysis of user preference evolution through time series network, further develop multimodal models to improve accuracy, and explore knowledge distillation technology to compress the volume, so as to provide a better solution for real-time analysis of the scene. These improvements will promote the fine-grained sentiment analysis technology to play a greater value in the catering e-commerce decision support system.

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